In traditional databases, data is often *normalized*, meaning it is spread across multiple tables and columns. This can make it difficult to search for data across multiple fields in a single query. Traditional databases are not designed to handle the types of unstructured and semi-structured data that are common in full-text search scenarios.

- *Text analysis and processing*—Search engines must often handle multiple languages and character sets, which traditional databases may not support. Search engines perform text analysis and processing to extract meaning from text, but traditional databases are not designed or optimized for this purpose.
- *Scalability and flexibility*—Full-text search engines are designed to handle large amounts of data and high query loads. Traditional databases can have scalability problems when dealing with large amounts of text data.

Search engines are designed from scratch to handle unstructured data, while databases are optimized for handling structured data. These limitations make traditional databases less suitable for use as full-text search engines; specialized search engine technologies such as Elasticsearch, Solr, Lucene, etc., are often used to provide advanced search functionality for text data.

NOTE Many databases have added text search capabilities to their feature sets. However, they still may not be able to deliver performance, scalability, and functionality on par with specialized full-text search engines.

Nothing is stopping us from embracing both worlds: in some use cases, a combination of traditional databases and search engines can be employed. For example, a database can be used for transactional purposes and a search engine for search and analytics. But our focus in this book is search engines—and Elasticsearch in particular. In the next section, we review the era of modern search engines before we introduce Elasticsearch.

1.3 Modern search engines

Modern search engines are trying hard to meet ever-growing business requirements by embracing new and exciting features every day. Cheap hardware combined with the explosion of data is leading to the emergence of these modern search beasts. Let's consider present-day search engines and the features and functionality they offer. We can summarize what a good modern search engine should provide as follows:

- First-class support for full-text (unstructured) and structured data
- Type-ahead suggestions, auto-correction, and "did-you-mean" recommendations
- Forgiveness for users' spelling mistakes
- Search capabilities on geolocations
- Easy scalability, either up or down, based on fluctuating demand
- Blazing performance: speedy indexing and search capabilities
- Architecture that provides a high-availability, fault-tolerant distributed system
- Support for machine learning functionalities